

ARIA: Adaptive Recurrent Intelligence Architecture

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Abstract

Catastrophic forgetting remains a fundamental obstacle in sequential learning. Most existing mitigations (EWC, replay buffers, progressive networks) protect weights without changing how the model allocates capacity across tasks. We introduce ARIA, an architecture in which capacity allocation is the learning signal. ARIA combines four jointly-trained mechanisms: (1) Morphogenic Attention (MA) -- heads split or merge via learned viability scores; (2) Plasticity-Gated MLP (PG-MLP) -- a dual fast/slow pathway with a learned per-token gate; (3) Architecture Genome Vector (AGV) -- a global latent vector conditioning all blocks via FiLM transforms; and (4) Cognitive Budget Allocator (CBA) -- per-layer compute budgets from input statistics. We introduce Slow-Pathway Consolidation (SPC), a Fisher regulariser applied only to slow-pathway weights. On Split-MNIST (5 seeds, parameter-matched baselines), ARIA+SPC achieves 98.57% average accuracy with 1.3% forgetting, outperforming EWC (97.35%, 2.3%) -- 43% less forgetting, 50% fewer Fisher parameters.

1. Introduction

A learning system encountering tasks sequentially faces the stability-plasticity dilemma. SGD-trained networks exhibit catastrophic forgetting. Regularisation (EWC, SI) is too conservative for long sequences; replay (DER++, ER) has memory cost; progressive architectures (PNN, PackNet) do not transfer. ARIA takes a fourth approach: the architecture adapts its structure, distributing capacity dynamically so old and new knowledge coexist. The core insight is that the fast/slow dichotomy of biological memory (Complementary Learning Systems) can be implemented as a learned per-neuron gate, with slow weights protected during high-plasticity phases and consolidated via targeted Fisher regularisation after each task.

Contributions:

- * Morphogenic Attention: first attention mechanism with dynamic head count via principled split/merge.
- * Plasticity-Gated MLP: dual fast/slow pathway with per-token learned gate and bimodal specialisation loss.
- * Slow-Pathway Consolidation (SPC): targeted Fisher on slow weights only -- 50% fewer params than EWC.
- * Architecture Genome Vector: latent z decoded to skip probs, attention temperature, FiLM scale/shift.
- * Cognitive Budget Allocator: per-layer compute from raw input statistics.

2. Method

2.1 Morphogenic Attention (MA)

Pre-allocates $H_{\max}$ heads with a boolean activation mask. Each head has a scalar viability updated by backprop. Every `morph_interval` steps: heads above the split threshold spawn a perturbed child; adjacent

heads above the merge threshold are averaged and one deactivated. A per-head cooldown prevents oscillation. Viabilities are softmax-normalised so output magnitude stays constant. FiLM conditioning (scale*h + shift, decoded from AGV) is applied to attention output.

2.2 Plasticity-Gated MLP (PG-MLP)

Dual fast and slow pathways gated by learned π in (0,1):

$$\text{output} = \pi * \text{fast_out}(\text{GeLU}(\text{fast_in}(x))) + (1-\pi) * \text{slow_out}(\text{GeLU}(\text{slow_in}(x)))$$

Bimodal loss $L_{\text{plastic}} = \lambda / (\text{mean}(\pi(1-\pi)) + \epsilon)$ pushes π toward 0 or 1, activated after warmup steps. Slow-pathway gradients multiplied by $(1 - \text{mean_}\pi)$ after each backward pass: high plasticity $\rightarrow$ slow path barely updated $\rightarrow$ consolidated knowledge protected.

2.3 Architecture Genome Vector (AGV)

Learnable z in $\mathbb{R}^{32}$, co-optimised with weights. Decoded to per-layer skip probabilities, attention temperature > 0.5 , FiLM scale near 1, FiLM shift near 0. Regularised by $\|z\|^2$. FiLM conditioning is stronger than additive injection because it rescales entire feature dimensions.

2.4 Cognitive Budget Allocator (CBA)

Predicts b in $[0,1]^L$ from $[\text{std}(x), \text{entropy}(x), \text{range}(x)]$. Block output: $b \cdot \text{full} + (1-b) \cdot \text{skip}$. Penalised by $\beta \cdot \text{mean}(b)$ to encourage sparse compute allocation.

2.5 Slow-Pathway Consolidation (SPC)

After task t , diagonal Fisher information estimated over slow weights only:

$$L_{\text{SPC}} = \lambda * \sum_{t' < t} \sum_k F_k(t') * (\theta_k - \theta_k^*)^2$$

Fast weights remain unconstrained. This resolves the EWC-PG-MLP conflict: EWC penalises fast weights from moving, reducing plasticity. SPC leaves them free. With $L=4$ layers, SPC uses 16 Fisher tensors vs 32+ for EWC (50% reduction in consolidation overhead).

3. Experiments

Benchmark: Split-MNIST -- 5 binary tasks on MNIST digit pairs (0/1, 2/3, 4/5, 6/7, 8/9), task-incremental setting, 784-dim input. All baselines parameter-matched to ARIA (~3.77M params). Optimiser: AdamW $\text{lr}=3\text{e-}4$, weight decay $1\text{e-}4$, gradient clip 1.0, 5 epochs/task, batch 64. Seeds: {42, 123, 999, 7, 2024}.

4. Results

ARIA+SPC achieves the best accuracy and lowest forgetting. Gap over EWC: +1.22pp accuracy, 0.96pp less forgetting (43% relative reduction). ARIA-noSPC already beats EWC by +1.08pp -- the architecture alone wins. SPC adds +0.14pp and halves forgetting variance (0.84% to 0.39%). DER++ underperforms due to the small replay buffer at matched parameter count.

Model	Avg Acc (up)	Forgetting (down)	BWT
ARIA+SPC (ours)	98.57 +/- 0.28%	1.33 +/- 0.39%	-1.32 +/- 0.40%
ARIA-noSPC (ours)	98.43 +/- 0.70%	1.63 +/- 0.84%	-1.63 +/- 0.85%
EWC	97.35 +/- 2.04%	2.29 +/- 2.78%	-2.25 +/- 2.77%
DER++	75.63 +/- 6.04%	30.18 +/- 7.55%	-30.18 +/- 7.55%
StaticMLP	76.91 +/- 3.02%	28.45 +/- 3.71%	-28.45 +/- 3.71%

Table 1: Split-MNIST (5 seeds). ARIA rows highlighted. All baselines parameter-matched (~3.77M params).

5. Analysis

5.1 Why SPC outperforms EWC

EWC regularises all weights uniformly, including fast-pathway weights designed to be volatile. This conflicts with PG-MLP: the plasticity gate routes new-task information through the fast pathway, but EWC penalises those weights from moving. SPC leaves fast weights unconstrained. ARIA-noSPC (no Fisher regularisation) already beats EWC by 1.08pp, confirming the architectural advantage precedes consolidation. SPC then adds consistent stability without interfering with plasticity.

5.2 The gradient dampening direction

The slow-pathway gradient multiplier must be $(1 - \text{mean_pi})$, not mean_pi . An earlier version used mean_pi (inverted), actively destabilising the slow path -- this caused pre-fix ablations to show ARIA-noPG outperforming ARIA-Full. The corrected implementation produces the results in Table 1.

5.3 Morphogenesis

Head counts remained at 4 across all 5 tasks -- the split threshold was not exceeded on Split-MNIST. The benchmark is too simple to require structural adaptation; PG-MLP + SPC suffice. Morphogenesis is expected to activate on harder, longer benchmarks (10+ tasks, CIFAR-100), planned for future work.

6. Conclusion

ARIA is a continual learning architecture that adapts its structure during training. Four jointly-trained mechanisms with Slow-Pathway Consolidation achieve 98.57% on Split-MNIST with 1.3% forgetting, outperforming parameter-matched EWC by 1.22pp and reducing forgetting by 43%. The corrected gradient dampening direction in PG-MLP was the critical enabling fix. Code: <https://github.com/rsd-darshan/ARIA>

References

- [1] Buzzega et al. (2020). Dark experience for general continual learning. NeurIPS.
- [2] Kirkpatrick et al. (2017). Overcoming catastrophic forgetting in neural networks. PNAS 114(13).
- [3] Lopez-Paz and Ranzato (2017). Gradient episodic memory for continual learning. NeurIPS.
- [4] Loshchilov and Hutter (2019). Decoupled weight decay regularization. ICLR.
- [5] McClelland et al. (1995). Why there are complementary learning systems. Psych. Review 102(3).
- [6] McCloskey and Cohen (1989). Catastrophic interference in connectionist networks. Psych. Learning.
- [7] Perez et al. (2018). FiLM: Visual reasoning with a general conditioning layer. AAAI.
- [8] Rusu et al. (2016). Progressive neural networks. arXiv:1606.04671.
- [9] Serra et al. (2018). Overcoming catastrophic forgetting with hard attention to the task. ICML.
- [10] Yoon et al. (2018). Lifelong learning with dynamically expandable networks. ICLR.
- [11] Zenke et al. (2017). Continual learning through synaptic intelligence. ICML.